SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

COURSE NAME:Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

18/20

Code of Conduct:

I Abangjithan S / 192521299 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: S. Ab

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations.**
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 - Video conferencing

TCP / UDP or Application Layer ?

—> The problem is mainly related to the transport protocol and application layer.

—> video conferencing needs low delay and low jitter.

—> UDP is generally better for real time video and audio.

—> TCP may cause delay because it retransmits lost packets.

—> 8% packet loss and 180ms jitter are causing poor quality

—> Improper application buffering can also cause video freezing.

2. Justification :

Real-time multi-media app require

—> Low latency (less than 150ms preferred)

—> Low Filter (less than 30ms)

—> Minimal packet loss (below 1%)

UDP is preferred because

—> No retransmission

—> Faster delivery

—> supports real-time voice and video.

3. Recommandation

—> continue using UDP

—> Enable RDP

—> use RTCP for network monitoring

Application Level improvements

—> use adaptive filter buffers

—> Apply adaptive bitrate streaming

—> Enable Forword error correcti

—> Increase bandwidth during peak ho

CASE STUDY 2 - DNS Infrastructure optimization

1. Reasons for high DNS resolution time

High DNS lookup time may occur because of

—> centralized DNS server

—> DNS server overload

—> cache misses

—> High Network congestion

—> poor TTL configuration

2 Proposed DNS Architecture

use :

—> Anycast DNS :

* Multiple DNS server shows some I

→ DNS Pre-fetching

* Frequently accessed domains are resolved before users request them.

→ TTL optimization

* Static records: TTL = 1-24 hours
* Dynamic records: TTL = 60-300 sec

Architecture:

Client
↓
Local DNS cache
↓
Anycast DNS Network
↓
Nearest DNS server
↓
Architective DNS

Advantages:

→ Faster DNS resolution
→ Better Scalability
→ High availability.
→ Automatic failover
→ Lower server load.

Limitation:

→ Higher deployment cost
→ cache inconsistency with very high TTL
→ Reauires global Anycast support

CASE STUDY 3 - Distributed DNS for cloud services :

1. Impact of centralized DNS

* A contracted DNS Architecture cause

→ High lookup latency.

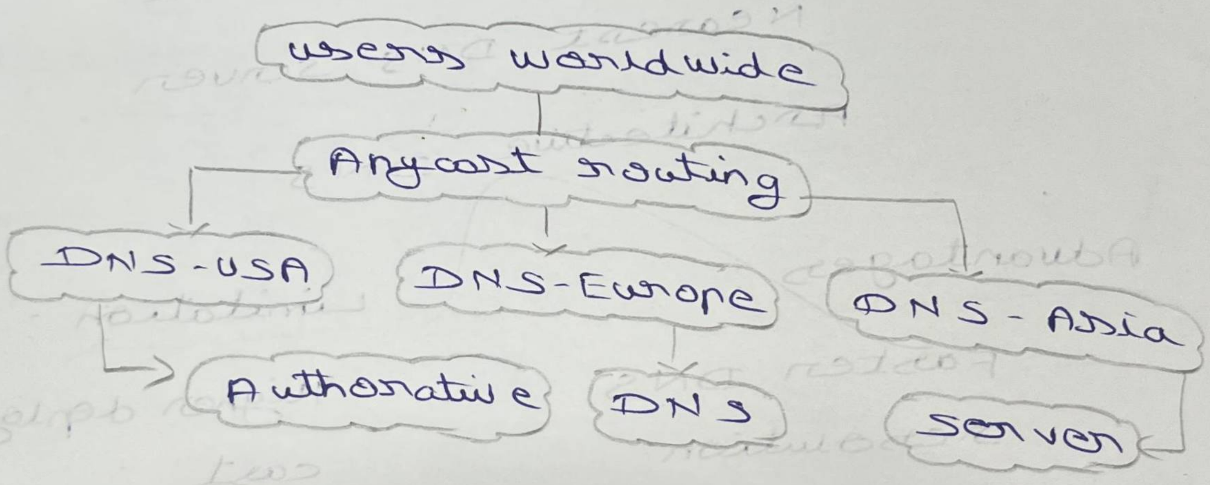
→ simple point of failure.

→ server overload.

Result :

→ DNS increases to around 800ms

2. Distributed DNS Solution



Features:

→ Multiple DNS server

→ Anycast routing.

→ DNS caching.

3. Inference of TTL, Anycast & DNS caching

High TTL

—> Less DNS traffic.

—> Slower updates.

Low TTL

—> Faster record updates.

—> More DNS queries.

DNS caching

—> Stores previous DNS response.

—> Reduces lookup time

CASE STUDY 4 - TCP Performance in stock trading platform.

1. Three TCP - related factors.

a) Flow control

—> Small receive window limits data transmissions.

—> causes delays.

b) Nagle's algorithm

—> Introduces additional delay.

—> unsuitable for high - frequency trade.

c) SYN Queue limitation.

—> New TCP connection wait or dropped

→ connection setup, time increase

2. Effect on transaction processing.

Factor	Effect
Flow control	Slower order process due to reduced transmission
Nagle's Algorithm	Delays sending small buy / sell orders.

3. TCP configuration recommendation

→ Disable Nagle's Algorithm.

→ Increase TCP receive / send buffer size.

→ Enable TCP fast open

→ Increase SYN backlog queue.

→ Reduce excessive buffering

→ Enable selective Acknowledgement (SACK).